SCL's Chip Finishing Activity Flow

Digital Design Division (DDD)
VLSI Design Group (VDG)
SCL

Dummy fill

- **Dummy fill can be done in either of two ways : Batch mode using header file or Calibre GUI.**
- **Procedure for Batch Mode is as follows :**
- **SCL had provided Dummy Fill Header file (~/scl180/pdk/cdns/sclpdk_v3/HOTCODE/techs/ts18scl/calibre/default_scr/headers/DUMMY.header).**
- **Procedure** to insert Dummy fill

1.User has to run

Calibre –drc Dummy.header

This will generate <top_with_dummy.gds>

2. Create new library in virtuoso with SCL Technology file.

3.Import <top_with_dummy.gds> in above library.

4. Place the top cell (same topcell name as original input) from above Library on the topcell of the original data.

5. Place both cells at origin and save with different cellname.

Dummy.header file (Sample)

```
// In order to run as stand alone (Not using SCL customization) you must define metal combination and select which layers to run on
// For metal combination -
// #define <one of the below metal combination>
// STd top metal:
//    4M1L,5M1L,6M1L
// layer selection options:
// #define MT_gen
// #define M5_gen
// #define M4_gen
// #define M3_gen
// #define M2_gen
// #define M1_gen
// #define AA_gen
// #define GC_gen

// Input GDS/Cellname/System Definition .
  LAYOUT PATH "layout_path.gds"
  LAYOUT PRIMARY "layout_top_cell"
  LAYOUT SYSTEM GDSII

// Output Database/report
  DRC SUMMARY REPORT "top_cell.db" REPLACE HIER
  DRC RESULTS DATABASE "top_cell.db" GDS2 // _dummy

  DRC MAXIMUM RESULTS ALL
  DRC MAXIMUM VERTEX ALL
  DRC CELL NAME YES CELL SPACE XFORM

INCLUDE "$RDS_TECH/$RDS_CDS_TECH/calibre/default_scr/DUMMYFILL_TS18SL_SCL_CALIBRE”
```

Note:

- User has to generate all dummy layers.
- However if he faces issue with specific layer , e.g. If GC layer violation, then he has to first run GC dummy only, then active and metal.
- Sometimes user has to run dummy fill multiple times according to violation type.
- Put no-dummy layer of required field (Active, GC and metals) on the portion where dummy fill not required.
- Make sure the part of layout where no dummy layer inserted has no DRC coverage area error.(if errors flags, make it error free by inserting dummy manually).

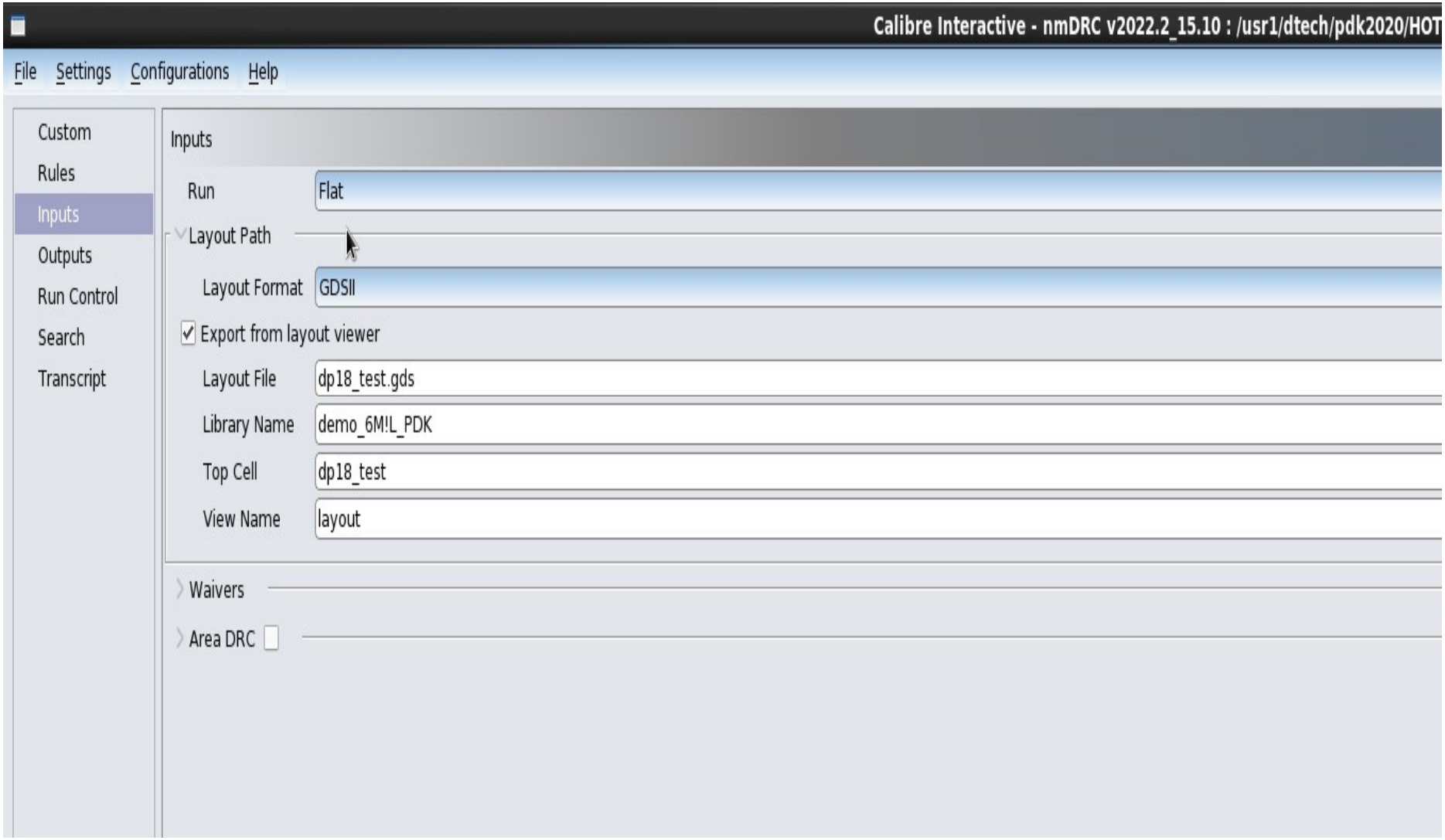
Dummy fill (Calibre GUI)

- **Procedure for dummy fill using Calibre GUI is as follows :**
- **Procedure** to insert Dummy fill

1. Open CalibreDRC
2. Under Custom Tab, Select DUMMY_FILL in DRC Run Options.
3. Under Inputs Tab, Select FLAT in Run Options.
4. Under Outputs Tab, Select GDSII as Format for dummy fill.

Dummy fill (Calibre GUI)

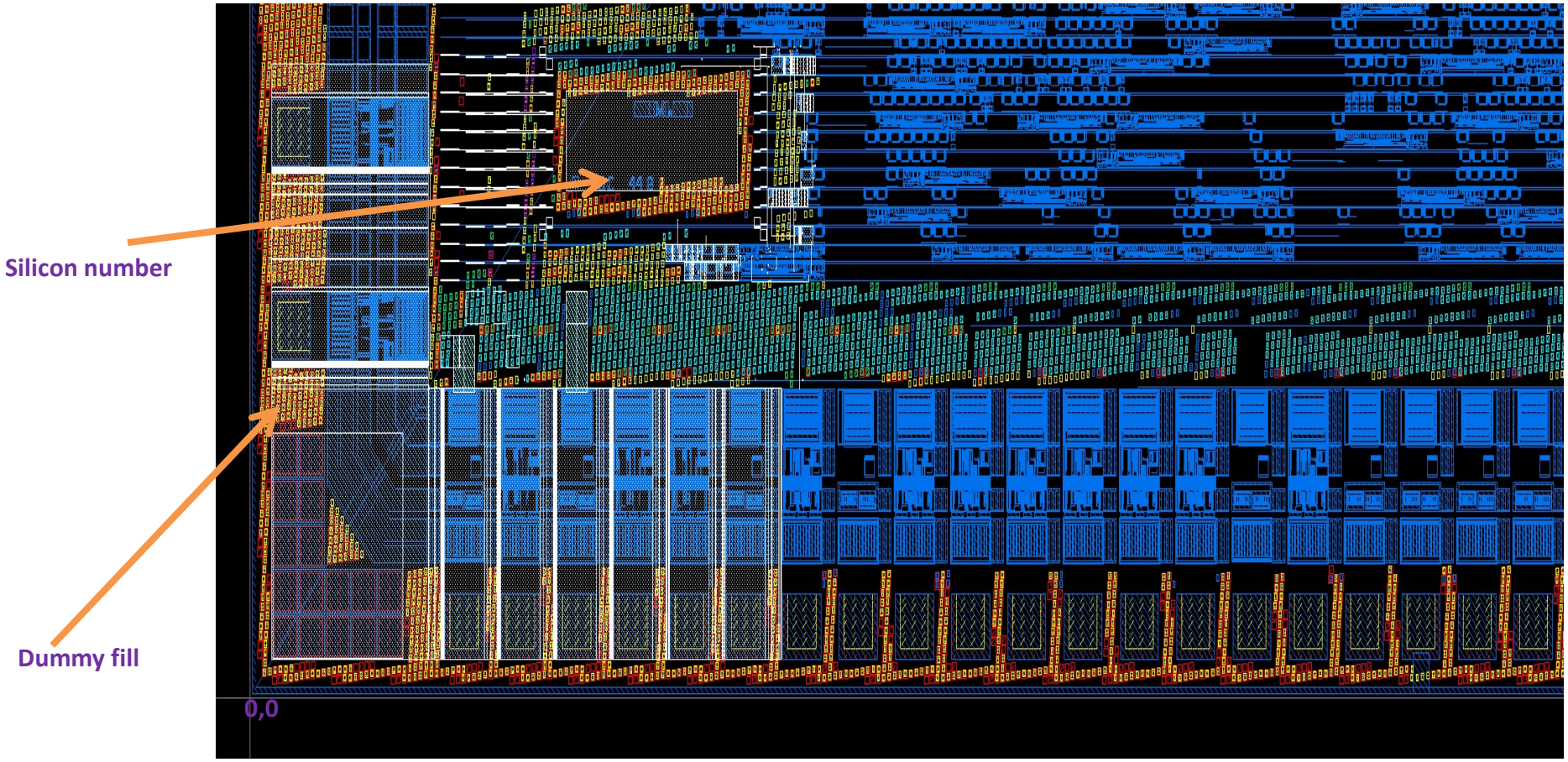
Dummy fill (Calibre GUI)



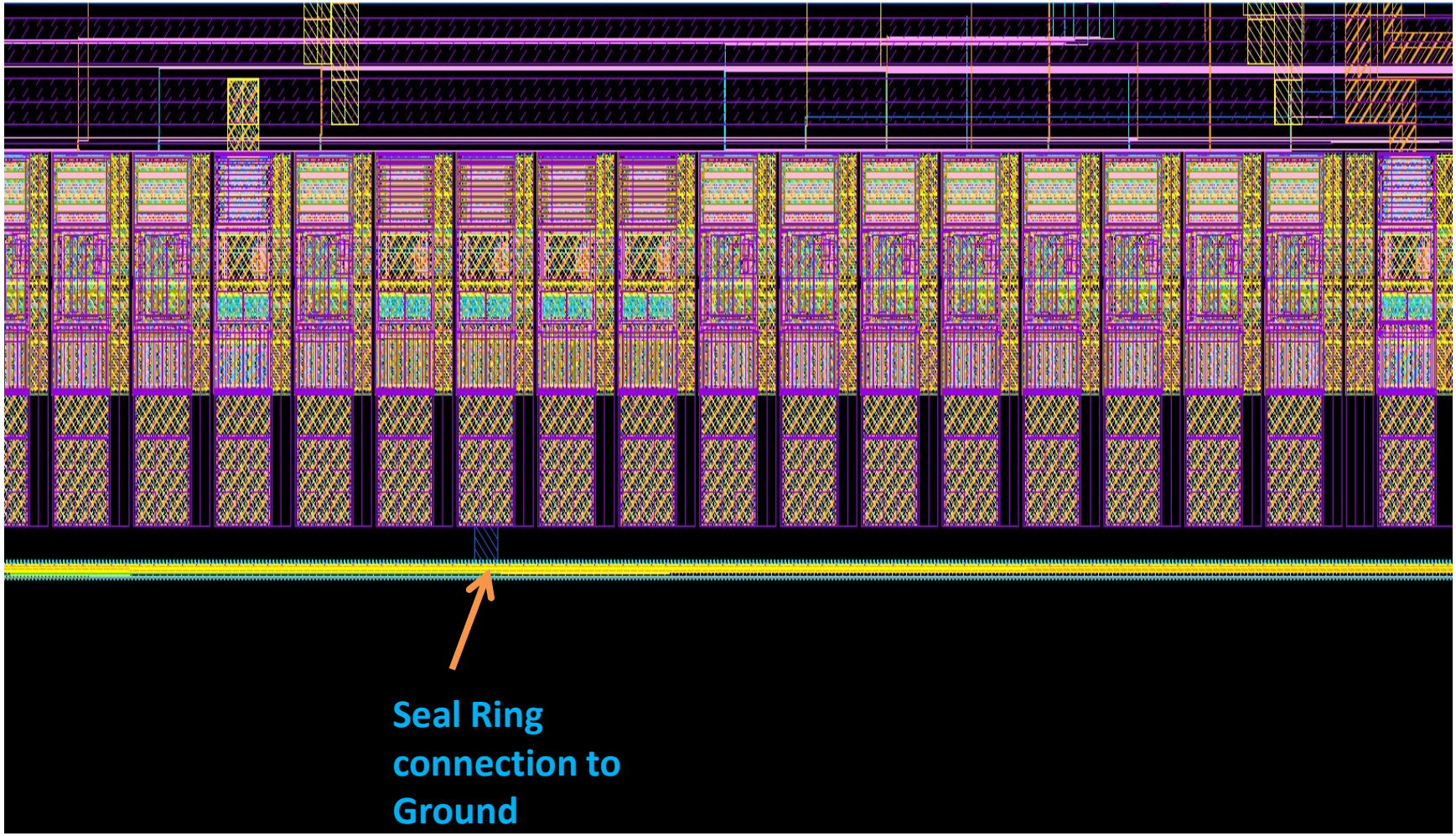
Dummy fill (Calibre GUI)

Seal ring and Silicon number

- SCL had provided Silicon number (default number) and seal ring GDS database along with procedure.
- User has to take Silicon number from SCL Process team (MASK team) as per SCL policy.



Seal Ring



Note: Seal ring should be connected to either to VSS or VSSO.